A PROOF OF THEOREM 5.1

In this section, we first provide essential lemmas and corollaries before presenting the full formal description of Theorem 5.1 and the proof of it.

Lemma A.1. For any $q \in Q, p \in P$, let $\theta(q,p) \in [0,\pi]$ denote the angle between them and $\mathrm{Ham}(\phi_i(q), \phi_i(p))$ the Hamming distance between their signatures, setting $k_{\mathrm{sim}} = O\left(\frac{\log(m/\delta)}{\varepsilon}\right) = O\left(\frac{\log(m)}{\varepsilon}\right)$, then with at least probability 1-δ:

$$\Pr\left(\left|\mathrm{Ham}(\phi_i(q), \phi_i(p)) - k_{sim} \cdot \frac{\theta(q,p)}{\pi}\right| > \sqrt{\varepsilon} k_{sim}\right) \leq \exp^{-2\epsilon k_{sim}} \leq \frac{\varepsilon\delta}{m^2}.$$

where m represents the sum of the number of vectors in one multi-vector query Q and one multi-vector document P.

The proof is given in [20].

Lemma A.2. Let $x_0, x_1, \ldots, x_k \in \mathbb{R}^d$ be unit vectors, and suppose that for some $\epsilon \in (0,1)$ the vectors $x_1, \ldots, x_k$ satisfy a *similarity ball condition*:

$$1 - \langle x_i, x_j \rangle \leq \epsilon, \quad \forall 1 \leq i, j \leq k.$$

Define the scores

$$s_i = \langle x_0, x_i \rangle, \quad i = 1, \ldots, k,$$

and let

$$s_{(1)} = \max_i s_i, \quad s_{(2)} = \text{second largest of } \{s_i\}.$$

Then:

(1) **Deterministic bound:**

$$\Delta := s_{(1)} - s_{(2)} \leq \epsilon.$$

(2) **Probabilistic refinement (sub-Gaussian extreme value scaling):**
If the deviations of s_i inside the similarity ball are modeled as independent sub-Gaussian variables of scale ϵ, i.e.,

$$s_i = \mu + z_i, \quad 0 \leq z_i \leq \epsilon, \quad z_i \text{ sub-Gaussian},$$

then for large k, with high probability,

$$\Delta = s_{(1)} - s_{(2)} \sim O\Big(\frac{\epsilon}{1 + \sqrt{\log k}}\Big).$$

Proof. **Step 1: Deterministic bound.**
For any pair $i, j \geq 1$,

$$|s_i - s_j| = |\langle x_0, x_i \rangle - \langle x_0, x_j \rangle| = |\langle x_0, x_i - x_j \rangle|.$$

Using the similarity ball condition:

$$1 - \langle x_i, x_j \rangle \leq \epsilon \quad \Longrightarrow \quad \langle x_i, x_j \rangle \geq 1 - \epsilon.$$

By the similarity-triangle inequality:

$$|\langle x_0, x_i - x_j \rangle| \leq 1 - \langle x_i, x_j \rangle \leq \epsilon.$$

Hence, for all i, j,

$$|s_i - s_j| \leq \epsilon.$$

Taking $s_{(1)} = \max_i s_i$ and $s_{(2)}$ = second largest:

$$\Delta = s_{(1)} - s_{(2)} \leq \epsilon.$$

Step 2: Probabilistic (EVT) refinement.
Model the scores as

$$s_i = \mu + z_i, \quad i = 1, \ldots, k,$$

where μ is a central reference in the similarity ball, and $z_i \in [0, \epsilon]$ are independent sub-Gaussian deviations.

Extreme value theory for sub-Gaussian variables [29] implies that for large k:

$$z_{(1)} - z_{(2)} \sim \frac{\sigma}{1 + \sqrt{\log k}},$$

where $\sigma = O(\epsilon)$ is the sub-Gaussian scale. Hence,

$$\Delta = s_{(1)} - s_{(2)} = z_{(1)} - z_{(2)} \sim O\Big(\frac{\epsilon}{1 + \sqrt{\log k}}\Big).$$

□

A.1 Formal description and proof of Theorem 5.1

Theorem (Approximation error of $\overline{\text{MaxSim}}_i$). *With the same choices of k, R and d_{proj} as used in Theorem 3.2, i.e., $k = O(\frac{\log(m)}{\varepsilon})$, $d_{\text{proj}} = O(\frac{\log(m/\epsilon)}{\epsilon^2})$, and $R = O(\frac{\log N}{\epsilon^2})$, the following inequality holds with high probability (i.e., $1 - 1/poly(N)$):*

$$0 \leq \frac{1}{|Q|}\left[\text{MaxSim}(Q,P) - \overline{\text{MaxSim}}_i(Q,P)\right] = O(\frac{1}{B}\sum_b \frac{\varepsilon}{1+\sqrt{\log n_{i,b}}}),$$

, where N is the number of documents, $n_{i,b}$ represents the number of document vectors from P assigned to the bucket b under the repetition i.

Proof. To analyze the gap between $\text{MaxSim}(Q,P)$ and $\overline{\text{MaxSim}}_i(Q,P)$, we perform this analysis for each query vector $q \in Q$. Hence, we analyze the gap between $m_i(q,P)$ and $\text{MaxSim}(q,P)$ first. The former one is defined in Section 5.1 while the latter one represents the similarity between q and its most similar document vector from P.

First, given a query vector q, suppose it is assigned to the bucket b, then two cases are considered here. The first case is that the ground-truth most similar document vector to q is also assigned to the bucket b. Hence, $\text{MaxSim}(Q,P) = \overline{\text{MaxSim}}_i(Q,P)$ so that the inequality to be proved holds. On the other hand, if the ground-truth most similar document vector to q is not assigned to the bucket b, then we define the following useful quantities:

$$P_{\text{bucket}} := \{p \in P : \phi_i(p) = b\}, \qquad p^* := \arg\max_{p\in P}\langle q,p\rangle, \qquad \mathcal{A}_i := P_{\text{bucket}} \cup \{p^*\},$$

let $\{X_{(1)}, X_{(2)}, \cdots\} = \{\langle q,p\rangle | p \in \mathcal{A}_i\}$, which are sorted as $X_{(1)} \geq X_{(2)} \geq \cdots$. Then

$$\text{MaxSim}(q,P) = \langle q,p^*\rangle := X_{(1)}, m_i(q,P) = X_{(2)}$$

Suppose $\{X_{(j)}\}$ follows the sub-Gaussian distribution, then we can directly apply Lemma A.2 to it, leading to

$$\mathbb{E}\Big[\text{MaxSim}(q,P) - m_i(q,P)\Big] = O(\frac{\varepsilon}{1+\sqrt{\log n_{i,b}}}).$$

By averaging across all buckets and all query vectors, the following formula holds:

$$\frac{1}{|Q|}\left[\text{MaxSim}(Q,P) - \overline{\text{MaxSim}}_i(Q,P)\right] = O(\frac{1}{|Q|}\sum_{b=\phi_i(q)} \frac{\varepsilon}{1+\sqrt{\log n_{i,b}}})$$

Considering that for an arbitrary multi-vector query Q, its query vector $q \in Q$ can be assigned into arbitrary buckets, we thus take expectations over all Q's, yielding,

$$\mathbb{E}_Q\Big[\frac{1}{|Q|}\left[\text{MaxSim}(Q,P) - \overline{\text{MaxSim}}_i(Q,P)\right]\Big] = O(\frac{1}{B}\sum_b \frac{\varepsilon}{1+\sqrt{\log n_{i,b}}})$$

Hence, $\frac{1}{|Q|}\left[\text{MaxSim}(Q,P) - \overline{\text{MaxSim}}_i(Q,P)\right] = O(\frac{1}{B}\sum_b \frac{\varepsilon}{1+\sqrt{\log n_{i,b}}})$ holds with high probability.

□

B PROOF OF THEOREM 5.2

Before presenting the proof of Theorem 5.2, we provide an important lemma.

B.1 An important lemma

Lemma B.1. Let $a_1, \ldots, a_n \in [-1,1]$. Then

$$\max_{j\in[n]} a_j \;-\; \frac{1}{n}\sum_{j=1}^{n} a_j \;\leq\; 2\Big(1-\frac{1}{n}\Big).$$

Proof. Let $a_{\max} = \max_j a_j \leq 1$. Since $a_j \geq -1$, $\frac{1}{n}\sum_{j=1}^n a_j \geq \frac{1}{n}a_{\max} + \frac{n-1}{n}(-1)$. Rearranging gives $a_{\max} - \frac{1}{n}\sum_j a_j \leq (1-\frac{1}{n})a_{\max} + \frac{n-1}{n} \leq 2(1-\frac{1}{n})$. □

B.2 Formal proof of Theorem 5.2

Proof. For a fixed query vector q, following the proof of Theorem 5.1, we employ P_{bucket} to denote the document vectors that are assigned to the same bucket as q or the ones belonging to the nearest bucket if the bucket $\phi_i(q)$ contains no document vectors. P_{bucket} are those document vectors that are compared against q in FDE computation. Hence, the similarity score between q and each vector in P_{bucket} could be regarded as a sequence between -1 and 1. Following the notations of Lemma B.1, we denote those similarity scores as $a_1, a_2, \ldots, a_n$, where $n = n_{i,b}$.

In addition, according to the definition of $m_i(q, P)$ (defined in Section 5.1), the maximal a_i among all $a_1, a_2, \ldots, a_n$ corresponds to $m_i(q, P)$. Likewise, we define $\text{FDE}_i(q, P)$ as the MaxSim score between one query vector q and the document P, which is equivalent to the average of them is equal to $\text{FDE}_i(q, P)$ according to its definition. Hence, according to Lemma B.1, the following inequality holds:

$$m_i(q, P) - FDE_i(q, P) \leq 2(1 - \frac{1}{n_{i,b}}). \tag{11}$$

Summing up the above formula across all q yields:

$$\overline{\text{MaxSim}}_i(Q, P) - FDE_i(Q, P) \leq \sum_{q, b=\phi_i(q)} 2(1 - \frac{1}{n_{i,b}}) = \sum_{b=1}^{B} q_{i,Q}(b)\Big(1 - \frac{1}{\max\{1, n_{i,b}\}}\Big), \tag{12}$$

in which, $q_{i,Q}(b)$ denotes the number of query vectors assigned to the bucket b.

Let $d_k := \max\{1, n_{i,P}(k)\} \geq 1$ and $q_k := q_{i,Q}(k)$. Using $\sum_{k=1}^{B} q_k = |Q|$, the upper bound of Equation (12) (denoted by $U_i(Q, P)$) is rewritten as

$$U_i(Q, P) = |Q| \cdot \Big(\frac{2}{|Q|} \sum_{k=1}^{B} q_k \Big(1 - \frac{1}{d_k}\Big)\Big) = |Q| \cdot \Big(2 - \frac{2}{|Q|} \sum_{k=1}^{B} \frac{q_k}{d_k}\Big).$$

Define weights $w_k := q_k/|Q|$, so $w_k \geq 0$ and $\sum_{k=1}^{B} w_k = 1$. Then

$$U_i(Q, P) = |Q| \cdot \Big(2 - 2 \sum_{k=1}^{B} \frac{w_k}{d_k}\Big).$$

Apply the Cauchy-Schwarz inequality:

$$\sum_{k=1}^{B} \frac{w_k}{d_k} = \sum_{k=1}^{B} \frac{w_k^2}{w_k d_k} \geq \frac{\frac{1}{B}(\sum_{k=1}^{B} w_k)^2}{\sum_{k=1}^{B} w_k d_k} = \frac{1}{B \sum_{k=1}^{B} w_k d_k}.$$

Since $\sum_{k=1}^{B} w_k d_k$ is a convex combination of $\{d_k\}$, we have $\sum_{k=1}^{B} w_k d_k \leq \max_k d_k = d_{i,P}^{\max}$. Therefore,

$$\sum_{k=1}^{B} \frac{w_k}{d_k} \geq \frac{1}{B \cdot d_{i,P}^{\max}} \quad \Longrightarrow \quad U_i(Q, P) \leq 2 - 2 \cdot \frac{1}{d_{i,P}^{\max}} = 2\Big(1 - \frac{1}{d_{i,P}^{\max}}\Big),$$

which completes the proof. □

C PROOF OF THEOREM 6.1

Proof. First of all, since we are deriving the stable rank of $\mathscr{P}^{(i,b)}$, normalizing each row vector does not impact the stable rank. Hence, we can assume that the norm of each row vector is 1. Plus, according to Claim A.4 in [20], the following formula holds with high probability for any pair of row vectors, x and y, which belong to the same bucket:

$$\cos(x, y) = \langle x, y \rangle = 1 - O(\epsilon) \tag{13}$$

According to the definition of stable rank [45], it is defined as the ratio between the Frobenius norm and Spectral norm as follows: $\text{srank}(\mathscr{P}^{(i,b)}) = \|\mathscr{P}^{(i,b)}\|_F^2 / \|\mathscr{P}^{(i,b)}\|_2^2$.

The Frobenius norm and Spectral norm of $\mathscr{P}^{(i,b)}$ is analyzed as follows:

(i) Frobenius norm. Because each row has unit norm,

$$\|\mathscr{P}^{(i,b)}\|_F^2 \;=\; \sum_{j=1}^{N} \|u_j\|_2^2 \;=\; N,$$

where u_j represents the j_{th} row of $\mathscr{P}^{(i,b)}$.

(ii) Spectral norm. According to the definition of the Spectral norm, the following inequality holds for any vector μ with L2 norm as 1:

$$\|\mathscr{P}^{(i,b)}\|_2^2 = \lambda_{\max}(\mathscr{P}^{(i,b)} \mathscr{P}^{(i,b)\top}) \geq \mu^\top \mathscr{P}^{(i,b)} \mathscr{P}^{(i,b)\top} \mu$$

By configuring $\mu = \frac{1}{\sqrt{N}}[1, 1, 1, \ldots, 1]$, the above formula is further derived as:

$$\|\mathscr{P}^{(i,b)}\|_2^2 \geq \mu^\top \mathscr{P}^{(i,b)} \mathscr{P}^{(i,b)\top} \mu = \frac{1}{N} \sum_{i,j} \mu_i^\top \mu_j$$

Equation (13) suggests that $\langle \mu_i^\top, \mu_j \rangle = 1 - O(\epsilon)$, which is incorporated to the above formula, yielding:

$$\|\mathscr{P}^{(i,b)}\|_2^2 \geq 1 + \frac{1}{N}(N^2 - N)(1 - O(\epsilon)) = N - (N - 1)O(\epsilon)$$

Combining (i) and (ii) yields

$$\text{srank}(\mathscr{P}^{(i,b)}) \;=\; \frac{\|\mathscr{P}^{(i,b)}\|_F^2}{\|\mathscr{P}^{(i,b)}\|_2^2} \;\leq\; \frac{N}{N-(N-1)O(\epsilon)} \;=\; 1+O(\epsilon).$$

□

D PROOF OF THEOREM 6.2

Proof. We prove zero means and zero covariance with high probability in what follows, respectively.

Proof of Zero Means. We first prove that $\mathbb{E}(\mathscr{P}^{(i)}) = \mathbb{E}(\mathscr{P}^{(j)}) = \mathbf{0}$ holds with high probability.

For the random transformation, $\psi_i(x) = \frac{1}{\sqrt{d_{\text{proj}}}} S_i x$, used for FDE vector generation, it contains i.i.d. Rademacher entries (± 1). Hence for every deterministic vector v,

$$\mathbb{E}_{\psi_i}[\psi_i(v)] = \mathbf{0},$$

Recall that for any multi-vector document X, $F_i(X)$ is defined as follows:

$$F_i(X) \;:=\; \Big(\psi_i\big(\tilde{p}_{i,1}(X)\big)^\top \;\|\; \cdots \;\|\; \psi_i\big(\tilde{p}_{i,B}(X)\big)^\top\Big)$$

Hence, the expectation of each $\psi_i\big(\tilde{p}_{i,1}(X)\big)$ over ϕ_i and ψ_i is derived as follows:

$$\mathbb{E}_{\phi_i,\psi_i}(\psi_i\big(\tilde{p}_{i,1}(X)\big)) = \mathbb{E}_{\psi_i}(\psi_i\big(\mathbb{E}_{\phi_i}(\tilde{p}_{i,1}(X))\big)),$$

which holds since $\psi_i(\cdot)$ is a linear transformation. Considering that $\mathbb{E}_{\phi_i}(\tilde{p}_{i,1}(X))\big)$ is independent of ψ_i, which can thus be viewed as a constant when taking expectations over ϕ_i, then the expectation of the above formula is $\mathbf{0}$. Incorporating this result into $F_i(x)$ yields,

$$\mathbb{E}_{\phi,\psi_i}(F_i(X)) \;:=\; \Big(\mathbb{E}_{\phi,\psi_i}(\psi_i\big(\tilde{p}_{i,1}(X)\big))^\top \;\|\; \cdots \;\|\; \mathbb{E}_{\phi,\psi_i}(\psi_i\big(\tilde{p}_{i,B}(X)\big))^\top\Big) = \Big(\mathbf{0}^\top\|\mathbf{0}^\top\|\cdots\|\mathbf{0}^\top\Big) = \mathbf{0}^\top$$

Considering that $\mathscr{P}^{(i)}$ is constructed by stacking all $F_i(x)$ across all documents, which are independent from each other, then $\mathbb{E}_{\phi_i,\psi_i}(\mathscr{P}^{(i)}) = \mathbf{0}$

Proof of Zero Covariance. We apply the law of total covariance by viewing (ψ_i, ψ_j) as a third random variable:

$$\text{Cov}(F_i(X), F_j(X)) = \mathbb{E}_{\psi_i,\psi_j}\Big[\text{Cov}(F_i(X), F_j(X) \mid (\psi_i,\psi_j))\Big] + \text{Cov}\Big(\mathbb{E}_{\psi_i,\psi_j}[F_i(X) \mid (\psi_i,\psi_j)],\, \mathbb{E}_{\psi_i,\psi_j}[F_j(X) \mid (\psi_i,\psi_j)]\Big).$$

Due to the independency between ψ_i and ψ_j, the above formula is further transformed into:

$$\text{Cov}(F_i(X), F_j(X)) = \mathbb{E}_{\psi_i,\psi_j}\Big[\text{Cov}(F_i(X), F_j(X) \mid (\psi_i,\psi_j))\Big] + \text{Cov}\Big(\mathbb{E}_{\psi_i}[F_i(X) \mid \psi_i],\, \mathbb{E}_{\psi_j}[F_j(X) \mid \psi_j]\Big).$$

We show both terms are zero.

By the zero-mean property of each $F_i(X)$ $\mathbb{E}_{\psi_j}[F_j(X) \mid \psi_j] = \mathbf{0}$. Similarly, $\mathbb{E}_{\psi_i}[F_i(X) \mid \psi_i] = \mathbf{0}$. Hence

$$\text{Cov}\Big(\mathbb{E}_{\psi_i}[F_i(X) \mid \psi_i],\, \mathbb{E}_{\psi_j}[F_j(X) \mid \psi_j]\Big) = \text{Cov}(\mathbf{0},\mathbf{0}) = \mathbf{0}.$$

By definition,

$$\mathbb{E}_{\psi_i,\psi_j}\Big[\text{Cov}(F_i(X), F_j(X) \mid (\psi_i,\psi_j))\Big] = \mathbb{E}_{\psi_i,\psi_j}\Big[F_i(X)F_j(X) \mid (\psi_i,\psi_j)\Big] - \mathbb{E}_{\psi_i}[F_i(X) \mid \psi_i]\cdot\mathbb{E}_{\psi_j}\Big[F_j(X) \mid \psi_j\Big].$$

The second term is $\mathbf{0}$ by Step 1. It remains to show $\mathbb{E}_{\psi_i,\psi_j}\Big[F_i(X)F_j(X) \mid (\psi_i,\psi_j)\Big] = \mathbf{0}$.

Since ψ_i and ψ_j are independent, we can apply the expectation over ψ_i and ψ_j one after another on the above formula, yielding $\mathbb{E}_{\psi_i}\Big[F_i(X)\cdot\mathbb{E}_{\psi_j}(F_j(X) \mid \psi_j) \mid (\psi_i,\psi_j)\Big] = \mathbf{0}$

□

E PROOF OF THEOREM 7.1

Proof. Let us analyze the eigen-vectors of each single-repetition single-block matrix, $\widehat{\mathscr{P}}^{(i,b)}$, which has dimensionality $N \times d_{\text{proj}}$. It can be further expressed as $\widehat{\mathscr{P}}^{(i,b)} = [\mu_1, \mu_2, \ldots, \mu_N]^\top$ following the notation in the proof of Theorem 6.1. In addition, according to Equation (13), due to the high pairwise similarity between vectors, we can rewrite each μ_i as $\mu_i = \alpha_i\mu + e_i$, where μ is a unit vector, e_i is a vector orthogonal to μ and $\|e_i\| = O(\sqrt{\epsilon})$. Incorporating this into $\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)}$ yields:

$$\mathscr{P}^{(i,b)\top}\mathscr{P}^{(i,b)} = [\mu_1, \mu_2, \ldots, \mu_N]\cdot[\mu_1, \mu_2, \ldots, \mu_N]^\top = \sum_j \mu_j\mu_j{}^\top = \sum_j \alpha_j^2\mu\mu^\top + E,$$

where $\|E\| = O(\epsilon)$. This indicates that $\mathscr{P}^{(i,b)\top}\mathscr{P}^{(i,b)}$ can be approximated by a rank-1 matrix. Likewise, for $\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)}$, it can be rewritten into a similar form:

$$\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)} = \sum_{j\in\widehat{\mathcal{P}}} \alpha_j^2\mu\mu^\top + E',$$

where $\|E'\| = O(\epsilon)$

The above derivation suggests that both $\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)}$ and $\mathscr{P}^{(i,b)\top}\mathscr{P}^{(i,b)}$ are dominant by one rank-1 matrix $\mu\mu^\top$ times some constant. Suppose when we perform sampling, we appropriately rescale $\widehat{\mathscr{P}}^{(i,b)}$ so that $\frac{1}{N^s-1}\sum_{j\in\widehat{\mathscr{P}}}\alpha_j^2$ has roughly the same norm as $\frac{1}{N-1}\sum_j \alpha_j$, in which N^s is the size of sampled document subset. As a result, $\|\frac{1}{N-1}\mathscr{P}^{(i,b)\top}\mathscr{P}^{(i,b)} - \frac{1}{N^s-1}\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)}\| = O(\epsilon)$.

Then, for the entire single-repetition matrix, $\mathscr{P}^{(i)}$ and $\widehat{\mathscr{P}}^{(i)}$, their covariance can be expanded as the following formulas using the above dominant eigen-vectors:

$$\begin{aligned}
\mathrm{Cov}(\mathscr{P}^{(i)}) &= \frac{1}{N-1}(\mathscr{P}^{(i)})^\top\mathscr{P}^{(i)} \\
&= \frac{1}{N-1}\begin{bmatrix} \mathscr{P}^{(i,1)\top}\mathscr{P}^{(i,1)} & \mathscr{P}^{(i,1)\top}\mathscr{P}^{(i,2)} & \dots & \mathscr{P}^{(i,1)\top}\mathscr{P}^{(i,B)} \\ \mathscr{P}^{(i,2)\top}\mathscr{P}^{(i,1)} & \mathscr{P}^{(i,2)\top}\mathscr{P}^{(i,2)} & \dots & \mathscr{P}^{(i,2)\top}\mathscr{P}^{(i,B)} \\ \dots & \dots & \dots & \dots \\ \mathscr{P}^{(i,B)\top}\mathscr{P}^{(i,1)} & \mathscr{P}^{(i,B)\top}\mathscr{P}^{(i,2)} & \dots & \mathscr{P}^{(i,B)\top}\mathscr{P}^{(i,B)} \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\mathrm{Cov}(\widehat{\mathscr{P}}^{(i)}) &= \frac{1}{N^s-1}(\widehat{\mathscr{P}}^{(i)})^\top\widehat{\mathscr{P}}^{(i)} \\
&= \frac{1}{N^s-1}\begin{bmatrix} \widehat{\mathscr{P}}^{(i,1)\top}\widehat{\mathscr{P}}^{(i,1)} & \widehat{\mathscr{P}}^{(i,1)\top}\widehat{\mathscr{P}}^{(i,2)} & \dots & \widehat{\mathscr{P}}^{(i,1)\top}\widehat{\mathscr{P}}^{(i,B)} \\ \widehat{\mathscr{P}}^{(i,2)\top}\widehat{\mathscr{P}}^{(i,1)} & \widehat{\mathscr{P}}^{(i,2)\top}\widehat{\mathscr{P}}^{(i,2)} & \dots & \widehat{\mathscr{P}}^{(i,2)\top}\widehat{\mathscr{P}}^{(i,B)} \\ \dots & \dots & \dots & \dots \\ \widehat{\mathscr{P}}^{(i,B)\top}\widehat{\mathscr{P}}^{(i,1)} & \widehat{\mathscr{P}}^{(i,B)\top}\widehat{\mathscr{P}}^{(i,2)} & \dots & \widehat{\mathscr{P}}^{(i,B)\top}\widehat{\mathscr{P}}^{(i,B)} \end{bmatrix}
\end{aligned}$$

Since $\|\frac{1}{N-1}\mathscr{P}^{(i,b)\top}\mathscr{P}^{(i,b)} - \frac{1}{N^s-1}\widehat{\mathscr{P}}^{(i,b)\top}\widehat{\mathscr{P}}^{(i,b)}\| = O(\epsilon)$, we can derive that $\|\mathrm{Cov}(\mathscr{P}^{(i)}) - \mathrm{Cov}(\widehat{\mathscr{P}}^{(i)})\|$ is also bounded by $O(\epsilon)$. By denoting all eigen-vectors of $\mathscr{P}^{(i)})^\top\mathscr{P}^{(i)}$ and $\widehat{\mathscr{P}}^{(i)})^\top\widehat{\mathscr{P}}^{(i)}$ are $V_1, V_2, \dots,$ and $\widehat{V}_1, \widehat{V}_2, \dots,$ and their corresponding eigen-values are $\lambda_1 > \lambda_2 > \dots$ and $\hat{\lambda}_1, \hat{\lambda}_2, \dots$. Then according to the Davis–Kahan theorem [10], if the gap between different eigen-values of $(\mathscr{P}^{(i)})^\top\mathscr{P}^{(i)}$ is large enough, then difference between V_i and $\widehat{V}_i$ is bounded by $O(\epsilon)$.

□

F PROOF OF THEOREM 8.1

Proof. Following the analysis in Appendix E, since $\mathscr{P}$ concatenates all single-repetition sub-matrices across all repetitions, all the eigen-vectors of $\mathscr{P}$ are thus $\{\{v^{(i,b)}\}_{b=1}^B\}_{i=1}^R$, which are collected from all buckets across all repetitions.

We then show that the FDE vector of any multi-vector query Q, represented by $F(Q) = [F_1(Q)\|F_2(Q)\|\cdots\|F_R(Q)]$, could be perfectly represented by the above eigen-vectors.

To begin with, we partition each $F_i(Q)$ into sub-vectors by buckets, denoted by $F_i(Q) = [F_i^{(1)}(Q) \parallel F_i^{(2)}(Q) \parallel \cdots \parallel F_i^{(B)}(Q)]$. Since $F_i^{(b)}(Q)$ lies in the same bucket as each row vector in $\mathscr{P}^{(i,b)}$, then the cosine similarity between $F_i^{(b)}(Q)$ and each row of $\mathscr{P}^{(i,b)}$ is lower bounded by $1 - O(\epsilon)$ according to Equation (13). Hence, following the analysis of Appendix E, we can rewrite $F_i^{(b)}(Q)$ as $F_i^{(b)}(Q) = \beta_i\mu + l_i$ where each row of $\mathscr{P}^{(i,b)}$ is also expressed by $\alpha_i\mu + e_i$, μ is one unit vector and l_i is one vector orthogonal to μ with norm $O(\sqrt{\epsilon})$.

By denoting the space spanned by the rows of $\mathscr{P}^{(i,b)}$ as $S^{(i,b)}$, the following inequality holds since the orthogonal projection of one vector on one space is the closest point in a subspace:

$$\|F_i^{(b)}(Q) - \mathrm{Proj}_{S^{(i,b)}}F_i^{(b)}(Q)\| \le \|F_i^{(b)}(Q) - \frac{\beta_i}{\alpha_1}\mu_1\| = \|e_i - \frac{\beta_i}{\alpha_i}l_i\| = O(\epsilon) \tag{14}$$

The above formula suggests that, the following equality holds:

$$F_i^{(b)}(Q) = x^{(i,b)\top}\cdot\mathscr{P}^{(i,b)} + E^{(i,b)} \tag{15}$$

where $x^{(i,b)} \in \mathbb{R}^{N\times 1}$ and $\|E^{(i,b)}\| = O(\epsilon)$.

Plus, the analysis in Appendix C suggests that the space spanned by the rows of $\mathscr{P}^{(i,b)}$, $S^{(i,b)}$ contributes only one dominant eigen-vector. In other words, $\mathscr{P}^{(i,b)}$ is a rank-1 matrix. Hence, $\mathscr{P}^{(i,b)}$ could be written as:

$$\mathscr{P}^{(i,b)} = a^{(i,b)}b^{(i,b)\top} + E^{(i,b)'}$$

where $a^{(i,b)} \in \mathbb{R}^{N\times 1}$ and $b^{(i,b)} \in \mathbb{R}^{1\times d_{\text{proj}}}$ and $\|E'\| = O(\epsilon)$. Incorporating the above formula into Equation (15) yields:

$$F_i^{(b)}(Q) = x^{(i,b)\top}\cdot a^{(i,b)}b^{(i,b)\top} + E^{(i,b)''} \tag{16}$$

where $E^{(i,b)''} = x^{(i,b)\top}\cdot E^{(i,b)'} + E^{(i,b)}$, which also satisfies $\|E^{(i,b)''}\| = O(\epsilon)$.

Since $F_i(Q)$ concatenates all $F_i^{(b)}(Q)$'s, incorporating the above formula into $F_i(Q)$ produces:

$$
\begin{aligned}
F_i(Q) &= [F_1^{(1)}(Q) \| F_1^{(2)}(Q) \| \cdots \| F_i^{(b)}(Q) \| \cdots] \\
&= [x^{(1,1)\top} \cdot a^{(1,1)} b^{(1,1)\top} + E^{(1,1)''} \parallel x^{(1,2)\top} \cdot a^{(1,2)} b^{(1,2)\top} + E^{(1,2)''} \parallel \cdots \| x^{(i,b)\top} \cdot a^{(i,b)} b^{(i,b)\top} + E^{(i,b)''} \parallel \cdots] \\
&= [x^{(1,1)\top} \cdot a^{(1,1)} b^{(1,1)\top} \parallel x^{(1,2)\top} \cdot a^{(1,2)} b^{(1,2)\top} \parallel \cdots \| x^{(i,b)\top} \cdot a^{(i,b)} b^{(i,b)\top} \parallel \cdots] + [E^{(1,1)''} \parallel E^{(1,2)''} \parallel \cdots \| E^{(i,b)''} \parallel \cdots] \\
&:= [c^{(1,1)} b^{(1,1)\top} \parallel c^{(1,2)} b^{(1,2)\top} \parallel \cdots \| c^{(i,b)} b^{(i,b)\top} \parallel \cdots] + [E^{(1,1)''} \parallel E^{(1,2)''} \parallel \cdots \| E^{(i,b)''} \parallel \cdots]
\end{aligned}
$$

in which, the norm of the residual, $[E^{(1,1)''} \parallel E^{(1,2)''} \parallel \cdots \| E^{(i,b)''} \parallel \cdots]$, is also bounded by $O(\epsilon)$. In addition, we represent the inner product $x^{(i,b)\top} \cdot a^{(i,b)}$ as $c^{(i,b)}$, a scalar value.

In contrast, we also concatenate all $\mathscr{P}^{(i,b)}$ in the same manner, leading to:

$$
\begin{aligned}
\mathscr{P} &= [\mathscr{P}^{(1,1)} \| \mathscr{P}^{(1,2)} \| \cdots \| \mathscr{P}^{(i,b)} \| \cdots] \\
&= [a^{(1,1)} b^{(1,1)\top} + E^{(1,1)'} \| a^{(1,2)} b^{(1,2)\top} + E^{(1,2)'} \| \cdots \| a^{(i,b)} b^{(i,b)\top} + E^{(i,b)'} \| \cdots],
\end{aligned}
$$

Then our goal is to identify one universal vector $x \in \mathbb{R}^N$ such that $F_i(Q) \approx x^\top \mathscr{P}$. To achieve this, we incorporate $x^\top \mathscr{P}$ into the above formula, leading to:

$$
x^\top \mathscr{P} = [x^\top a^{(1,1)} b^{(1,1)\top} + x^\top E^{(1,1)'} \| x^\top a^{(1,2)} b^{(1,2)\top} + x^\top E^{(1,2)'} \| \cdots \| x^\top a^{(i,b)} b^{(i,b)\top} + x^\top E^{(i,b)'} \| \cdots],
$$

in which, $x^\top a^{(i,b)}$ is expected to be equal to $c^{(i,b)}$. This suggests that we need to solve the following linear system:

$$
x^\top [a^{(1,1)}, a^{(1,2)}, \ldots, a^{(i,b)}, \ldots, a^{(R,B)}] = [c^{(1,1)}, c^{(1,2)}, \ldots, c^{(i,b)}, \ldots, c^{(R,B)}]
$$

Since N is typically far larger than $R \cdot B$, then the above linear system is an underdetermined system [49]. For such systems, we can choose the minimal-norm solution via pseudo-inverse, i.e.,:

$$
x = (AA^\top)^{-1} A \cdot [c^{(1,1)}, c^{(1,2)}, \ldots, c^{(i,b)}, \ldots, c^{(R,B)}]^\top,
$$

where $A = [a^{(1,1)}, a^{(1,2)}, \ldots, a^{(i,b)}, \ldots, a^{(R,B)}]$. This guarantees that the reconstruction error, $\|x^\top A - [c^{(1,1)}, c^{(1,2)}, \ldots, c^{(i,b)}, \ldots, c^{(R,B)}]\|$, is exactly zero [49]. This suggests that we can find such a x that can ensure that $F_i(Q) \approx x^\top \mathscr{P}$ and their gap is $O(\epsilon)$.

□